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Complementary Split-Prime Transfer Obstruction

Asymptotic injectivity for $P_3(x,y) = (x-y)^3 - 3(x+y)$

This report was prepared completely offline. It uses the five frozen facts in the research checkpoint, the already-audited construction for

$$R(x,y) = (x-y)^2 + 5(x+y),$$

and the standard results stated in Section I. No external verification is claimed.

A. Frozen checkpoint

The following inputs are used without re-proving them.

1. The totally real bounded-root-discriminant tower and its completely split auxiliary primes.
2. The Minkowski-box construction of the quadratic witness, including its lattice counts and the estimate
$$|R(A_j, A_j)| \le \Theta_R^{n_j} |A_j|^2, \qquad \Theta_R < 1.$$
3. The bridge identity
$$5P_3(x,y) = 5(x-y)^3 + 3(x-y)^2 - 3R(x,y).$$
4. Injectivity of the cubic map inside every exact quadratic-output fiber.
5. Maximal failure of transfer for the previously treated primes $q \equiv 5 \pmod{12}$.

Only the complementary choice $q \equiv 1 \pmod{12}$ is analyzed below.

B. Verdict: Outcome III

The Complementary Root Overlap Conjecture is false for the explicit $T = Q/8$ quadratic witness. In fact, the opposite phenomenon holds: the cubic map is asymptotically injective.

Main theorem

Let K be totally real of degree n, with $\mathrm{rd}(K) \le D$. Let

$$Q = \prod_{q \in \mathcal{S}} q, \qquad |\mathcal{S}| = t,$$

where the rational primes in $\mathcal{S}$ are distinct, satisfy $q \equiv 1 \pmod{12}$, and split completely in K. Define

$$\mathcal{B} = B_K(Q/8) = \{b \in \mathcal{O}_K : |\sigma(b)| \le Q/8 \text{ for every real embedding } \sigma\}, \qquad N = |\mathcal{B}|,$$

and, for one real embedding σ_0,

$$A = \left\{ \frac{\sigma_0(b)}{Q} : b \in \mathcal{B} \right\}.$$

Put

$$\delta = \frac{1}{2}\log Q - t\log 4 - \log(32\sqrt{D}). \tag{B.1}$$

If $\delta > 0$, then

$$\boxed{|P_3(A,A)| \ge \frac{|A|^2}{1+e^{-\delta n}}.} \tag{B.2}$$

For every successful complementary-prime quadratic witness with $T = Q/8$, the condition $\Theta_R < 1$ automatically implies $\delta > 0$. More explicitly, it implies $t \ge 8$, and one may replace δ in (B.2) by

$$\boxed{\delta_*(t) = \frac{1}{2}\log\left(\frac{41t!}{256(8/3)^t}\right) > 0.} \tag{B.3}$$

Consequently, along the tower $n = n_j \to \infty$,

$$1 \le M_{n_j} := \frac{|A_j|^2}{|P_3(A_j,A_j)|} \le 1 + e^{-\delta_* n_j},$$

and hence

$$\boxed{\lim_{j\to\infty} \frac{1}{n_j}\log\frac{|A_j|^2}{|P_3(A_j,A_j)|} = 0.} \tag{B.4}$$

This is the requested Outcome III. It closes the remaining transfer branch of the successful quadratic split-prime architecture. It does **not** disprove CPSI for other constructions.

C. The witness and the exact second-digit condition

For the supplied tower, fix the auxiliary parameter first. Its totally real layers K_j have

$$n_j = [K_j : \mathbb{Q}] \longrightarrow \infty, \qquad \mathrm{rd}(K_j) \le D,$$

where D is fixed. The split primes may be selected with $q \equiv 1 \pmod{12}$ by applying Chebotarev to the fixed compositum that also contains $\mathbb{Q}(i, \zeta_3)$. Once the primes, Q, and $T = Q/8$ are fixed, put

$$\mathcal{B}_j = B_{K_j}(T), \qquad A_j = \sigma_j(\mathcal{B}_j)/Q.$$

The frozen lattice estimates give

$$\left(\frac{Q}{8\sqrt{D}}\right)^{n_j} \le |A_j| \le \left(\frac{Q}{4} + 1\right)^{n_j}. \tag{C.1}$$

In particular, the elements of A_j are distinct and $|A_j| \to \infty$ once $Q/(8\sqrt{D}) > 1$.

For $b, c \in \mathcal{B}_j$, write

$$d = b - c, \qquad s = b + c,$$

and clear denominators:

$$S = d^2 + 5Qs = Q^2 R(b/Q, c/Q), \qquad Z = d^3 - 3Q^2 s = Q^3 P_3(b/Q, c/Q). \tag{C.2}$$

The bridge becomes

$$5Z = 5d^3 + 3Qd^2 - 3QS. \tag{C.3}$$

Thus the sets $\mathcal{Z}_S$ in CRO are precisely the values $5Z$ above the exact quadratic label S. Multiplication by 5 does not change cardinality.

If two ordered pairs give the same cubic output, and their differences and sums are (d, s) and (e, s'), then

$$d^3 - e^3 = 3Q^2(s - s'). \tag{C.4}$$

The essential point is therefore

$$\boxed{d^3 \equiv e^3 \pmod{Q^2 \mathcal{O}_K},} \tag{C.5}$$

not merely congruence modulo Q. The second q-adic digit is what makes almost all formal cube-root alternatives miss the narrow Minkowski window.

D. Main lemma

The new mechanism is total-positivity packing.

Let

$$\mathcal{D} = B_K(Q/4)$$

and define the off-diagonal cube-congruence set

$$\mathcal{C}_Q = \{(d, e) \in \mathcal{D}^2 : d \neq e,\ d^3 \equiv e^3 \pmod{Q^2\mathcal{O}_K}\}.$$

The central estimate is

$$\boxed{|\mathcal{C}_Q| \leq (4^{t+1}\sqrt{Q})^n.} \tag{D.1}$$

This estimate includes both unit cube-root branches and singular coordinates where both differences vanish modulo a split prime.

D.1. Ideal-coset box packing

We first record an elementary lemma.

Lemma D.1. If $\mathfrak{a} \subseteq \mathcal{O}_K$ is a nonzero integral ideal of norm A_0, then every coset $z + \mathfrak{a}$ satisfies

$$|(z + \mathfrak{a}) \cap B_K(L)| \leq \left(1 + \frac{2L}{A_0^{1/n}}\right)^n. \tag{D.2}$$

Proof. Partition each interval $[-L, L]$ into at most $1 + 2L/A_0^{1/n}$ half-open intervals of length strictly less than $A_0^{1/n}$. Two distinct points of the same product cell would differ by a nonzero $\alpha \in \mathfrak{a}$ with

$$|N_{K/\mathbb{Q}}(\alpha)| < A_0.$$

But $(\alpha) \subseteq \mathfrak{a}$, so $|N(\alpha)| \geq N(\mathfrak{a}) = A_0$, a contradiction. □

D.2. Four local types modulo $\mathfrak{p}^2$

Because every $q \mid Q$ splits completely,

$$Q^2\mathcal{O}_K = \prod_{\mathfrak{p} \mid Q} \mathfrak{p}^2,$$

and there are tn prime ideals in this product. Each has $N\mathfrak{p} = q$, and $\mathcal{O}_K/\mathfrak{p}^2 \cong \mathbb{Z}/q^2\mathbb{Z}$.

Fix a primitive cube root $\zeta_\mathfrak{p}$ modulo $\mathfrak{p}^2$. Hensel's lemma applies because $q \neq 3$: the three simple roots of $X^3 - 1$ modulo $\mathfrak{p}$ lift uniquely modulo $\mathfrak{p}^2$. Every local solution of $d^3 \equiv e^3 \pmod{\mathfrak{p}^2}$ has exactly one of four types:

$$\begin{array}{c|l} T & d \notin \mathfrak{p},\ e \equiv d \pmod{\mathfrak{p}^2}, \\ N_+ & d \notin \mathfrak{p},\ e \equiv \zeta_\mathfrak{p} d \pmod{\mathfrak{p}^2}, \\ N_- & d \notin \mathfrak{p},\ e \equiv \zeta_\mathfrak{p}^{-1} d \pmod{\mathfrak{p}^2}, \\ Z & d, e \in \mathfrak{p}. \end{array} \tag{D.3}$$

Indeed, if d is a unit, then e/d is one of the three cube roots of unity. If $d \in \mathfrak{p}$, the congruence forces $e \in \mathfrak{p}$, and then both cubes vanish modulo $\mathfrak{p}^2$.

A global type τ chooses one of these four alternatives at all tn split prime ideals. Let W_T, W_N, W_Z be the sums of $\log N\mathfrak{p}$ over the corresponding positions, combining N_+ and N_- in W_N. Then

$$W_T + W_N + W_Z = n \log Q. \tag{D.4}$$

Put

$$u = d - e, \qquad v = d + e,$$

and define

$$\mathfrak{J}_\tau = \prod_T \mathfrak{p}^2 \prod_Z \mathfrak{p}, \qquad \mathfrak{I}_\tau = \prod_{N_+ \cup N_-} \mathfrak{p}^2 \prod_Z \mathfrak{p}. \tag{D.5}$$

Then $u \in \mathfrak{J}_\tau$. Once u is fixed, v lies in one coset modulo $\mathfrak{I}_\tau$. At a nontrivial root position,

$$v \equiv \frac{1+\zeta_\mathfrak{p}}{1-\zeta_\mathfrak{p}} u \pmod{\mathfrak{p}^2}, \tag{D.6}$$

where $1 - \zeta_\mathfrak{p}$ is a unit; at a Z-position, $u, v \in \mathfrak{p}$. No global cube root of unity in K is being assumed: (D.6) is a separate local congruence at each $\mathfrak{p}$.

Write

$$a = \frac{2W_T + W_Z}{n}, \qquad b = \frac{2W_N + W_Z}{n}. \tag{D.7}$$

Then

$$N\mathfrak{J}_\tau = e^{an}, \qquad N\mathfrak{I}_\tau = e^{bn}, \qquad a + b = 2 \log Q. \tag{D.8}$$

Since $d, e \in B_K(Q/4)$, both u and v belong to $B_K(Q/2)$. Lemma D.1, first for u and then for the single v-coset, gives for the off-diagonal points of type τ

$$M_\tau \le [(1 + Qe^{-a})(1 + Qe^{-b})]^n. \tag{D.9}$$

D.3. The two packing regimes

Assume first that

$$a \ge \frac{1}{2} \log Q.$$

If the type contains an off-diagonal point, then $0 \ne u \in \mathfrak{J}_\tau$, so

$$e^{an} = N\mathfrak{J}_\tau \le |N_{K/\mathbb{Q}}(u)| \le (Q/2)^n. \tag{D.10}$$

For $x = Qe^{-a}$, this gives $2 \le x \le \sqrt{Q}$, while $Qe^{-b} = x^{-1}$. Hence

$$M_\tau \le (2 + x + x^{-1})^n \le (4\sqrt{Q})^n. \tag{D.11}$$

Now assume

$$a < \frac{1}{2}\log Q.$$

Let

$$h = \frac{W_N + W_Z}{n}.$$

Because $a \geq 2W_T/n$, (D.4) gives

$$h = \log Q - W_T/n > \frac{3}{4}\log Q. \tag{D.12}$$

Take two distinct off-diagonal points of the same global type, and denote their difference by $(\Delta d, \Delta e)$. At an N_+ or N_- position the same lifted root is used by both points, so $\Delta e \equiv \zeta^{\pm 1}\Delta d \pmod{\mathfrak{p}^2}$. At a Z-position both differences lie in $\mathfrak{p}$. Therefore

$$H(\Delta d, \Delta e) := (\Delta d)^2 + \Delta d\,\Delta e + (\Delta e)^2 \in \prod_{N_+ \cup N_- \cup Z} \mathfrak{p}^2. \tag{D.13}$$

The norm of the ideal on the right is e^{2nh}. The binary form

$$H(X, Y) = X^2 + XY + Y^2$$

is positive definite at every real embedding. For a nonzero pair, $H(\Delta d, \Delta e) \neq 0$, and ideal divisibility followed by arithmetic-geometric mean gives

$$\sum_{\sigma} H(\sigma\Delta d, \sigma\Delta e) \geq n e^{2h}. \tag{D.14}$$

Since

$$H(x, y) \leq \frac{3}{2}(x^2 + y^2),$$

distinct points of this type are separated in $\mathbb{R}^{2n}$ by at least

$$r = \sqrt{\frac{2n}{3}}e^h. \tag{D.15}$$

All type points lie in a Euclidean ball of radius

$$R = Q\sqrt{n/8}.$$

The disjoint-ball packing bound in dimension $2n$ yields

$$M_\tau \leq (1 + 2R/r)^{2n} \leq \left(1 + \frac{\sqrt{3}}{2}Q^{1/4}\right)^{2n} \leq (4\sqrt{Q})^n. \tag{D.16}$$

There are at most 4^{tn} global types. Summing (D.11) and (D.16) proves (D.1).

E. Complete collision-energy proof

Define

$$Z(b,c) = (b-c)^3 - 3Q^2(b+c)$$

on $\mathcal{B}^2$, and let

$$E_Z = \#\{(b,c,b',c') \in \mathcal{B}^4 : Z(b,c) = Z(b',c')\}.$$

For a collision put $d = b - c$ and $e = b' - c'$. Equation (C.4) puts every off-diagonal (d,e) in $\mathcal{C}_Q$.

If $d = e$, equality of outputs forces $b + c = b' + c'$, so the two ordered input pairs coincide. These contribute exactly N^2 to E_Z.

Fix $(d,e) \in \mathcal{C}_Q$. There are at most N possible first pairs (b,c) with difference d, because choosing $b \in \mathcal{B}$ determines $c = b - d$. Once the first pair is fixed, equality determines

$$s' = s + \frac{e^3 - d^3}{3Q^2}$$

and then uniquely determines

$$b' = \frac{s' + e}{2}, \qquad c' = \frac{s' - e}{2}.$$

These elements may fail to be integral or to lie in $\mathcal{B}$, but that only decreases the count. Therefore

$$E_Z \le N^2 + N|\mathcal{C}_Q|. \tag{E.1}$$

The frozen Blichfeldt estimate gives

$$N \ge \left(\frac{Q}{8\sqrt{D}}\right)^n. \tag{E.2}$$

Combining (D.1), (E.1), and (E.2), and putting

$$\lambda = \frac{32\sqrt{D}\,4^t}{\sqrt{Q}} = e^{-\delta}, \tag{E.3}$$

gives

$$E_Z \le N^2(1 + \lambda^n). \tag{E.4}$$

If $m(z)$ is the number of ordered input pairs producing z, then

$$\sum_z m(z) = N^2, \qquad \sum_z m(z)^2 = E_Z.$$

Cauchy-Schwarz gives

$$N^4 \le |Z(\mathcal{B},\mathcal{B})| E_Z,$$

so (E.4) proves (B.2). Because σ_0 is an injective field embedding and

$$P_3(\sigma_0(b)/Q, \sigma_0(c)/Q) = \sigma_0(Z(b,c))/Q^3,$$

the algebraic and real image cardinalities agree.

There is also a stronger incidence interpretation. From

$$\sum_z m(z)(m(z)-1) \le \lambda^n N^2$$

it follows that all but at most $\lambda^n N^2$ ordered input pairs lie in singleton cubic fibers. Thus the theorem is not merely a lower bound obtained from a large diagonal term: global cross-fiber collisions occupy an exponentially vanishing fraction of the Cartesian square.

F. Why quadratic success forces $\delta > 0$

For the complementary-prime quadratic witness, put

$$\rho = \prod_{q|Q} \frac{q+1}{2q}.$$

The frozen quadratic estimate at $T = Q/8$ is

$$|R(A,A)| \le \Theta_R^n |A|^2, \qquad \Theta_R = D\rho\left(164 + \frac{64}{Q}\right). \tag{F.1}$$

Suppose $\Theta_R < 1$. Since $\rho > 2^{-t}$ and $D \ge 1$,

$$1 > \Theta_R > 164D, 2^{-t}.$$

Consequently

$$t \ge 8, \qquad D < \frac{2^t}{164}. \tag{F.2}$$

Order the selected primes increasingly. The i-th distinct prime congruent to $1 \pmod{12}$ is larger than $12i$, whether or not every number in that progression is prime. Hence

$$Q > 12^t t!. \tag{F.3}$$

Using (F.2) and (F.3) in (E.3),

$$\lambda^2 = \frac{1024D\,16^t}{Q} < \frac{256}{41}\frac{(8/3)^t}{t!}. \tag{F.4}$$

At $t = 8$, the right side is less than 0.397. Its ratio at $t+1$ to its value at t is

$$\frac{8/3}{t+1} < 1,$$

so it decreases for every $t \ge 8$. Thus $\lambda < 1$, and (B.3) follows. Numerically, $\delta_*(8) > 0.463$.

This argument is stronger than merely choosing a favorable large-prime subfamily: every $q \equiv 1 \pmod{12}$, $T = Q/8$ witness that achieves the frozen quadratic inequality $\Theta_R < 1$ automatically lies in the near-injective cubic regime.

G. Gain-loss ledger and CRO

Quantity	Bound or rate	Role
Field degree	$n = n_j \to \infty$	Only growing parameter after the witness is fixed
Root discriminant	$\mathrm{rd}(K_j) \le D$	Fixed lattice cost
Input size	$(Q/(8\sqrt{D}))^n \le \lvert A_j\rvert \le (Q/4+1)^n$	Exponential growth, distinct real inputs
Quadratic image	$\lvert R(A_j, A_j)\rvert \le \Theta_R^n \lvert A_j\rvert^2$, $\Theta_R < 1$	Frozen benchmark success
Local Q^2 types	At most 4^{tn}	T, N_+, N_-, Z at every split coordinate
One type	At most $(4\sqrt{Q})^n$ difference pairs	Ideal packing or total positivity
All off-diagonal difference pairs	At most $(4^{t+1}\sqrt{Q})^n$	Main new lemma
Anchors over fixed (d, e)	At most $\lvert A_j\rvert$	Equality determines the second pair
Off-diagonal energy ratio	At most $\lambda^n = e^{-\delta n}$	Exponentially negligible cross-fiber overlap
Cubic image	At least $\lvert A_j\rvert^2/(1+e^{-\delta n})$	Asymptotic injectivity
GAIN rate	$n^{-1}\log M_n \to 0$	Rules out CRO and a power saving on this witness

For the exact quadratic fibers in the question,

$$\sum_S |\mathcal{Z}_S| = |A_j|^2, \qquad \left|\bigcup_S \mathcal{Z}_S\right| = |P_3(A_j, A_j)|.$$

Therefore

$$\boxed{\frac{1}{1+e^{-\delta n_j}} \le \frac{|\bigcup_S \mathcal{Z}_S|}{\sum_S |\mathcal{Z}_S|} \le 1.} \tag{G.1}$$

CRO requires the middle ratio to be at most $e^{-\eta n_j}$ for some $\eta > 0$. Equation (G.1) shows that it instead tends to 1.

The surviving μ_3-ambiguity is therefore formal local entropy, not realized global multiplicity. Exact equality demands a lift modulo Q^2; the product of the narrow archimedean window with ideal-norm and total-positivity separation leaves exponentially few nontrivial lifted partners.

Since $|A_j| \to \infty$, (B.2) also shows that for every fixed $c > 0$,

$$|P_3(A_j, A_j)| > |A_j|^{2-c}$$

for all sufficiently large j. Thus these quadratic witness sets cannot prove CPSI.

H. Consequence and exactly one next unresolved statement

Both congruence choices available in the fixed $T = Q/8$ quadratic split-prime transfer are now closed:

- the $q \equiv 5 \pmod{12}$ witness is exactly injective for P_3;
- the $q \equiv 1 \pmod{12}$ witness is asymptotically injective by (B.2).

The conclusion is about this transfer architecture, not about the polynomial itself. The one next problem is therefore the direct form of CPSI:

$$\boxed{\text{Construct explicit } A_j \subset \mathbb{R},\ |A_j| \to \infty,\ \text{such that } \liminf_{j\to\infty} \frac{\log(|A_j|^2/|P_3(A_j, A_j)|)}{\log |A_j|} > 0.}$$

Such a construction must introduce a source of cubic collisions not inherited from the quadratic square-residue witness.

I. Standard results used

For completeness, the proof invokes the following standard statements in these precise forms.

1. **Chebotarev density theorem.** For a finite Galois extension $L/\mathbb{Q}$ and a conjugacy class $C \subseteq \mathrm{Gal}(L/\mathbb{Q})$, the unramified rational primes with Frobenius class C have natural density $|C|/|\mathrm{Gal}(L/\mathbb{Q})|$. Here it supplies arbitrarily many primes split in the fixed tower quotient and in $\mathbb{Q}(i, \zeta_3)$.
2. **Blichfeldt's lattice principle.** If a measurable set in $\mathbb{R}^n$ has volume V and a full lattice has covolume Δ, some translate contains at least V/Δ lattice points. Applied to a cube and followed by subtraction, it gives (E.2).
3. **Hensel's simple-root lemma.** A root a of f modulo a prime ideal with $f'(a)$ invertible lifts uniquely modulo every higher power; only the lift from $\mathfrak{p}$ to $\mathfrak{p}^2$ is used here.
4. **Ideal norm divisibility.** If $0 \neq \alpha \in \mathfrak{a}$ for an integral ideal $\mathfrak{a}$, then $(\alpha) \subseteq \mathfrak{a}$ and $|N_{K/\mathbb{Q}}(\alpha)| \geq N\mathfrak{a}$.
5. **Arithmetic-geometric mean.** For positive real numbers $x_1, \ldots, x_n$, $n^{-1} \sum x_i \geq (\prod x_i)^{1/n}$.
6. **Euclidean packing bound.** If points in a ball of radius R in $\mathbb{R}^m$ are mutually at distance at least r, disjoint balls of radius $r/2$ give at most $(1 + 2R/r)^m$ points.
7. **Cauchy-Schwarz for fibers.** If a finite map has domain size M, image size L, and collision energy E, then $M^2 \leq LE$.